

Module 3 Lesson 4

Systems of Linear Equations in Two Variables (7.3)

Objectives

In this lesson, we will learn how to

- decide whether an ordered pair is a solution of a linear system;
- solve linear systems by graphing;
- solve linear systems by substitution;
- solve linear systems by addition;
- identify systems that do not have exactly one ordered-pair solution;
- solve problems using systems of linear equations.

1. Systems of Linear Equations

Two linear equations considered together are called a system of linear equations, or a linear system.

A solution to a system of linear equations in two variables is an ordered pair that satisfies both equations.

Solution of a Linear System

An ordered pair (x, y) is a solution of a linear system if it makes every equation in the system true.

For example, a system may look like

$$\begin{cases} 2x - 3y = -4, \\ 2x + y = 4. \end{cases}$$

To check whether an ordered pair is a solution, substitute the x -value and y -value into both equations.

2. Checking an Ordered Pair

Example 1: Determine whether $(1, 2)$ is a solution of the system:

$$\begin{cases} 2x - 3y = -4, \\ 2x + y = 4. \end{cases}$$

Solution

Because $(1, 2)$ has $x = 1$ and $y = 2$, substitute these values into both equations.

First equation:

$$2x - 3y = -4$$

$$2(1) - 3(2) = -4$$

$$2 - 6 = -4$$

$$-4 = -4.$$

This is true.

Second equation:

$$2x + y = 4$$

$$2(1) + 2 = 4$$

$$2 + 2 = 4$$

$$4 = 4.$$

This is true.

Therefore, $(1, 2)$ is a solution of the system.

3. Solving Linear Systems by Graphing

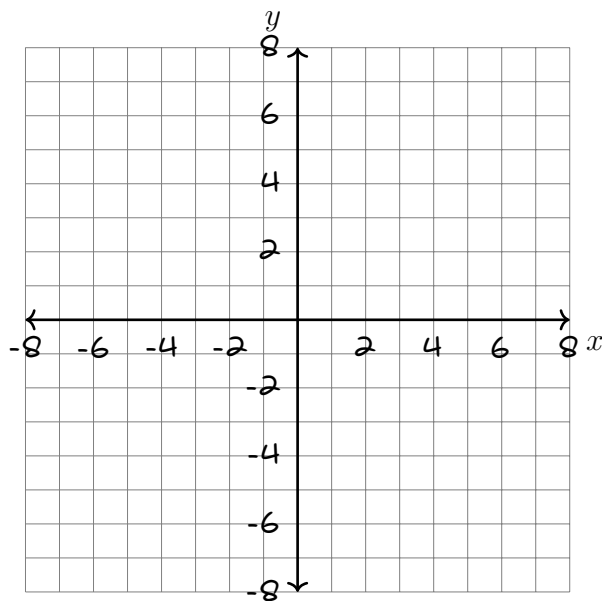
A solution to a system of two linear equations is the point where the two lines intersect.

Graphical Meaning

For a system with one solution, the coordinates of the point of intersection are the solution of the system.

Example 2: Solve the system by graphing:

$$\begin{cases} x + 2y = 2, \\ x - 2y = 6. \end{cases}$$



Solution

Graph the first equation:

$$x + 2y = 2.$$

Find intercepts.

For the x -intercept, set $y = 0$:

$$x + 2(0) = 2$$

$$x = 2.$$

So the first line passes through

$$(2, 0).$$

For the y -intercept, set $x = 0$:

$$0 + 2y = 2$$

$$y = 1.$$

So the first line passes through

$$(0, 1).$$

Graph the second equation:

$$x - 2y = 6.$$

For the x -intercept, set $y = 0$:

$$x - 2(0) = 6$$

$$x = 6.$$

So the second line passes through

$$(6, 0).$$

For the y -intercept, set $x = 0$:

$$0 - 2y = 6$$

$$y = -3.$$

So the second line passes through

$$(0, -3).$$

4. Solving Linear Systems by Substitution

The substitution method is useful when one equation is already solved for one variable.

Substitution Method

To solve a linear system by substitution:

1. Solve one equation for one variable in terms of the other variable.
2. Substitute that expression into the other equation.
3. Solve the resulting equation in one variable.
4. Back-substitute to find the value of the other variable.
5. Check the proposed solution in both original equations.

Example 3: Solve by the substitution method:

$$\begin{cases} y = -x - 1, \\ 4x - 3y = 24. \end{cases}$$

Solution

The first equation is already solved for y :

$$y = -x - 1.$$

Substitute $-x - 1$ for y in the second equation:

$$4x - 3y = 24$$

$$4x - 3(-x - 1) = 24.$$

Simplify:

$$4x + 3x + 3 = 24$$

$$7x + 3 = 24$$

$$7x = 21$$

$$x = 3.$$

Back-substitute $x = 3$ into

$$y = -x - 1.$$

$$y = -3 - 1 = -4.$$

The solution is

$$(3, -4).$$

5. Solving Linear Systems by Addition

The addition method is also called the elimination method.

The goal is to add the equations so that one variable is eliminated.

Addition Method

To solve a linear system by addition:

1. Rewrite both equations in the form

$$Ax + By = C.$$

2. If necessary, multiply one or both equations by nonzero numbers so that one variable will cancel.
3. Add the equations.
4. Solve the resulting equation in one variable.
5. Back-substitute to find the other variable.
6. Check the solution in both original equations.

Example 4: Solve by the addition method:

$$\begin{cases} 3x + 2y = 48, \\ 9x - 8y = -24. \end{cases}$$

Solution

Both equations are already in the form

$$Ax + By = C.$$

Multiply the first equation by -3 :

$$-3(3x + 2y = 48)$$

$$-9x - 6y = -144.$$

Keep the second equation unchanged:

$$9x - 8y = -24.$$

Now add:

$$\begin{array}{r} -9x - 6y = -144 \\ 9x - 8y = -24 \\ \hline -14y = -168 \end{array}$$

Solve:

$$y = 12.$$

Back-substitute into

$$3x + 2y = 48.$$

$$3x + 2(12) = 48$$

$$3x + 24 = 48$$

$$3x = 24$$

$$x = 8.$$

The solution is

$$(8, 12).$$

6. Types of Solutions for Linear Systems

A system of two linear equations in two variables may have:

- exactly one solution;
- no solution;
- infinitely many solutions.

Number of Solutions

Number of Solutions	Graphical Meaning
Exactly one solution	The two lines intersect at one point.
No solution	The two lines are parallel.
Infinitely many solutions	The two lines are identical.

7. A System with No Solution

Example 5: Solve the system:

$$\begin{cases} 4x + 6y = 12, \\ 6x + 9y = 12. \end{cases}$$

Solution

Use the addition method.
Multiply the first equation by 3:

$$12x + 18y = 36.$$

Multiply the second equation by -2 :

$$-12x - 18y = -24.$$

Add:

$$\begin{array}{r} 12x + 18y = 36 \\ -12x - 18y = -24 \\ \hline 0 = 12 \end{array}$$

The statement

$$0 = 12$$

is false.

Therefore, the system has no solution.

The solution set is

$$\emptyset.$$

8. A System with Infinitely Many Solutions

Example 6: Solve the system:

$$\begin{cases} y = 3x - 2, \\ 15x - 5y = 10. \end{cases}$$

Solution

Because the first equation is already solved for y , use substitution. Substitute $3x - 2$ for y in the second equation:

$$15x - 5y = 10$$

$$15x - 5(3x - 2) = 10.$$

Simplify:

$$15x - 15x + 10 = 10.$$

$$10 = 10.$$

This statement is true.

Therefore, the system has infinitely many solutions.

The two equations represent the same line.

9. When Both Variables Are Eliminated

Sometimes both variables disappear while solving a system.

Interpreting the Result

If both variables are eliminated:

- If the resulting statement is false, the system has no solution.
- If the resulting statement is true, the system has infinitely many solutions.

10. Modeling with Systems of Equations

Systems of equations can be used to model business situations.

A company may compare revenue and cost to determine when it breaks even.

Revenue and Cost Functions

If a company produces and sells x units of a product, then

$$R(x) = (\text{price per unit sold})x.$$

The cost function is

$$C(x) = \text{fixed cost} + (\text{cost per unit produced})x.$$

The point of intersection of the revenue and cost functions is called the break-even point.

At the break-even point,

$$R(x) = C(x).$$

11. Finding a Break-Even Point

Example 7: A company is planning to manufacture specialized wheelchairs.

The fixed cost will be \$500,000, and it will cost \$400 to produce each wheelchair.

Each wheelchair will be sold for \$600.

- Write the cost function $C(x)$.
- Write the revenue function $R(x)$.
- Determine the break-even point.
- Describe what the break-even point means.

Solution

The fixed cost is

$$500,000.$$

The variable cost is

$$400x.$$

So the cost function is

$$C(x) = 500,000 + 400x.$$

The company earns \$600 for each wheelchair sold, so the revenue function is

$$R(x) = 600x.$$

At the break-even point,

$$R(x) = C(x).$$

So,

$$600x = 500,000 + 400x.$$

Subtract $400x$ from both sides:

$$200x = 500,000.$$

Divide by 200:

$$x = 2500.$$

Now find the revenue or cost at $x = 2500$:

$$R(2500) = 600(2500) = 1,500,000.$$

The break-even point is

$$(2500, 1,500,000).$$

This means the company breaks even if it produces and sells 2500 wheelchairs for a total revenue of \$1,500,000.

12. Profit Function

Profit is revenue minus cost.

Profit Function

The profit generated after producing and selling x units of a product is

$$P(x) = R(x) - C(x),$$

where $R(x)$ is revenue and $C(x)$ is cost.

For the wheelchair example,

$$R(x) = 600x$$

and

$$C(x) = 500,000 + 400x.$$

So,

$$P(x) = R(x) - C(x)$$

$$P(x) = 600x - (500,000 + 400x)$$

$$P(x) = 200x - 500,000.$$

Example 8: Using the profit function

$$P(x) = 200x - 500,000,$$

find the profit when the company produces and sells

3000

wheelchairs.

Solution

Substitute $x = 3000$:

$$P(3000) = 200(3000) - 500,000.$$

$$P(3000) = 600,000 - 500,000.$$

$$P(3000) = 100,000.$$

The profit is

\$100,000.

13. More Practice

Example 9: Determine whether $(2, 1)$ is a solution of the system:

$$\begin{cases} 3x + y = 7, \\ x - 2y = 0. \end{cases}$$

Example 10: Solve by substitution:

$$\begin{cases} y = 2x + 1, \\ 3x + y = 16. \end{cases}$$

Example 11: Solve by addition:

$$\begin{cases} 2x + 3y = 13, \\ 4x - 3y = 5. \end{cases}$$

Summary

Key Ideas

- A system of linear equations consists of two or more linear equations.
- A solution of a system in two variables is an ordered pair that satisfies both equations.
- Graphically, the solution is the point where the lines intersect.
- The substitution method replaces one variable with an equivalent expression.
- The addition method eliminates one variable by adding equations.
- A system can have exactly one solution, no solution, or infinitely many solutions.
- No solution means the lines are parallel.
- Infinitely many solutions means the two equations represent the same line.
- Revenue, cost, and profit functions can be modeled using systems of equations.